



Running the 2008 Cell study



Running the 2014 Cell study

HELIOFLUX'S APPROACH

Research-stage hardware and software

Measure the light. Look for unique patterns.

Q1: CAPTURE

QSense



Concept art only

Sensitive enough to count individual photons

A photon-counting sensor built to record light in tissues is already going off. Nothing is shown in, repeated or drawn. The sample sits in the dark and the detector counts.

A closer look at the light.

Two longpass filters could not separate the groups. Filtering the light by wavelength could.

Q2: Compare cell samples

Cancer and non-cancer lines, each recorded in triplicate.

Q3: Filter the light

One wavelength band per recording.

Q4: Record the light

A photon-counter under the dish for a day

Color and timing: what's the difference?

Q5: INTERPRET

LuminAI



Concept art only

Built around the information in the signal

Software that learns the patterns in color and timing which, in this case, set cancer cells apart. It is not being trained, and every new dataset sharpens it.

WHAT DOES "PASSIVE MEASUREMENT" MEAN?

Dr. Nirrosha Mangan has published on this signal in cell cultures, in a mouse cancer model, and in living people. Each study answered one question and set up the next.

FOUNDATIONAL RESEARCH

2018

Cells in a dish

Cancer and non-cancer cells glow off different light.

2020

Cancer and healthy adults

In mice with melanoma, the signal shifted within 24 hours.

2021

Healthy adults

First light recorded on the head, alongside EEG.

2022

People with cancer

Light readings matched to biopsy results.

A foundation in published science.

Can the light tell cancer cells from non-cancer cells?

Yes, once it was **filtered**. At three wavelengths, cancer cells and non-cancer cells give off measurably different light. The best classification reached 92%.

Study details and original paper

2020

Cell research

Does the signal show up in a living animal?

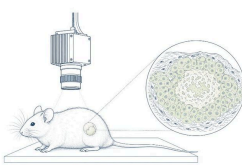
Yes, in 13 cell lines, the rhythm of the light separated cancer from non-cancer with 90% accuracy. In mice injected with melanoma, the signal shifted within 24 hours, long before a tumor could be felt.

Study details and original paper

HELIOFLUX · ASSETS/NEXT/REVIEW/STYLE-PACK/

timeline-full-full.jpg · 1200 × 13333 · in three strips, read left to right

VIEW THE FULL DOCUMENT



Photon images

Researchers watched the tumor's faint emission.

[Scientific source ↗](#)

Tissue composition

Emission patterns were linked with growth and viable tissue.

What the study shows

Findings correct & original papers

[Explore study ➔](#)

2009

VIEW THE FULL DOCUMENT

The human body's faint light is photographed

A highly sensitive camera photographed bare light from five healthy men. The signal varied across the body and through the day, landing its peak in late afternoon. It was too faint to see with the naked eye.

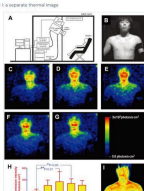
KEY TAKEAWAYS

- Direct human measurement, with images, experimental conditions and quantitative analysis available for reproduction.

ORIGINAL RESEARCH PAPER

C-D: photon intensity across the day. E: separate thermal image

FIGURE 1



Narrow photon emission changes through the day, as the human body's photon response revealed a signal that varied with time of day (**Figure 1**).

Digital aspect ↗

© 2009 American Nuclear Society | DOI

What the study shows

Findings correct & original papers

[Explore study ➔](#)

2009-2022

Reading the patterns

The question shifts to measuring what the light's colors and rhythm reveal about stress, cell chemistry and cancer in the lab.

KEY TAKEAWAYS

- It indicates if any

2011

REVIEW

Changing the chemistry changes the light

In mice, increasing oxidation strengthened the light, while removing oxygen reduced it. It separates human-like experiments, particularly in how oxygen can reduce emissions, linking the signal to specific chemical reactions.

[Download full paper ➔](#)

2014

REVIEW

A review brings together 56 human studies

A review gathered 56 studies measuring near-light in people and human samples. It mapped a growing research field and gaps in how studies were run rather than validating a particular medical test.

[Download full paper ➔](#)

2014

CELL STUDY **METHODS REVIEW** **CELL CULTURE**

Stressed cells show a changing light pattern

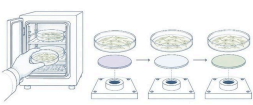
Doyle, Hargrett and colleagues measured near-light responses in cultures for 24 hours through a series of wavelength filters. Different colors of light peaked at different times, and the light appears, and other color is it, carries more information than how bright it is.

KEY TAKEAWAYS

- Coordinated photon measurements with changing culture conditions and modulated wavelength-released research.

ILLUSTRATION OF THE EXPERIMENT

2004



Across 24 hours

Different wavelength bands produced at different times.

[Scientific source ↗](#)

Beyond brightness

Which light was measured mattered too.

What the study shows

Findings correct & original papers

[Explore study ➔](#)

2015

REVIEW

A review separates measurements from bigger claims

This review found biological light emission well-established, but no reliable evidence in the studies reviewed for stronger claims about coordinated light waves or unusual quantum behavior.

KEY TAKEAWAYS

- An essential practical argument presenting all biophoton hypotheses as equally established science.

ILLUSTRATION OF THE REVIEW

2004

Detecting photons



Coherence hypothesis



Faint light exists

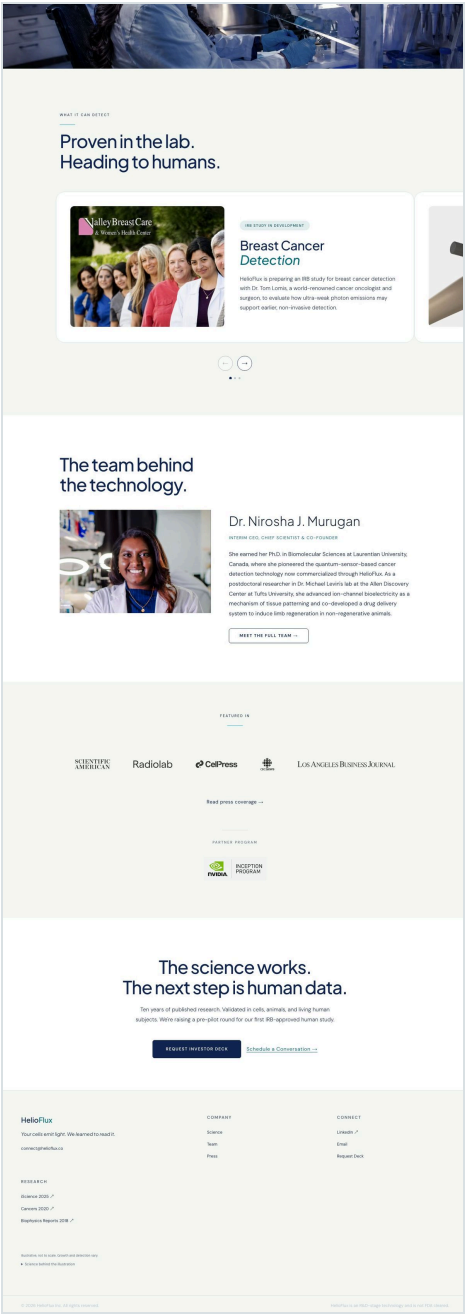
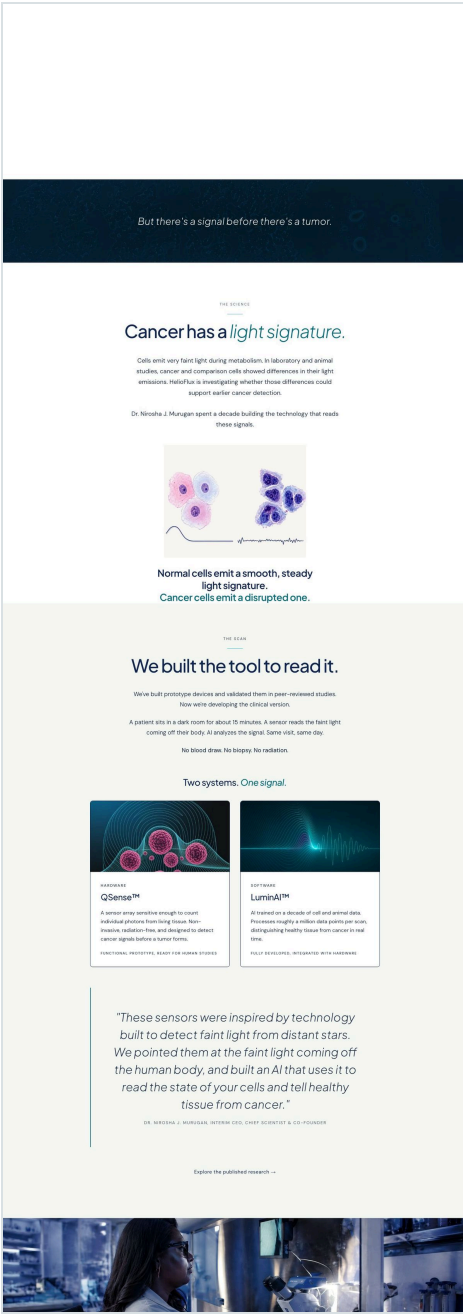
Suggested by experiments.

Coordinated light waves

Not reliably demonstrated in the research reviewed.

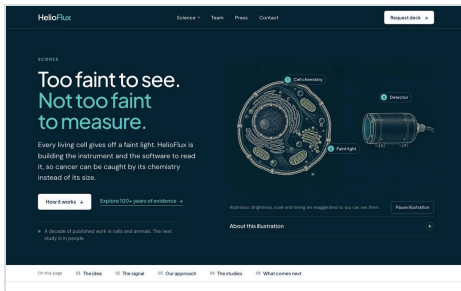
Home, the whole page

home-full-full.jpg · 1200 × 10158 · in three strips, read left to right



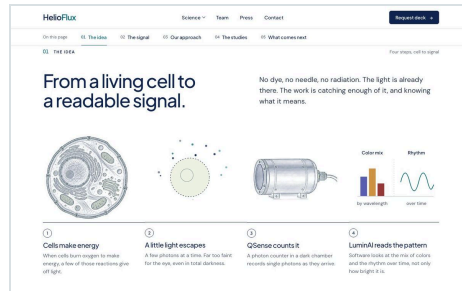
Screens and phone views

7 desktop screens at 1440 × 900 · 3 phone screens at 390 × 844



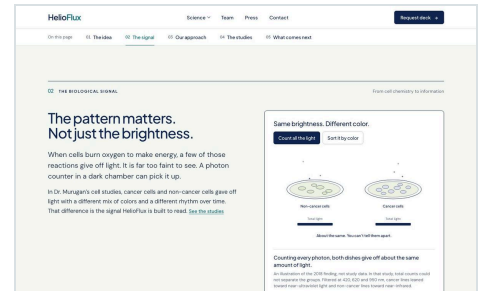
science-screen-00.jpg

1200 × 750



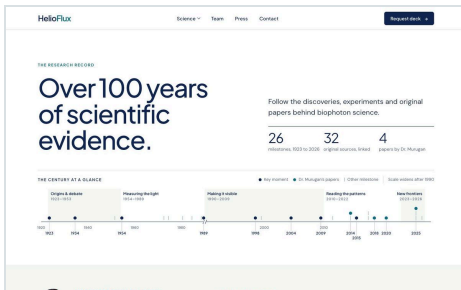
science-screen-01.jpg

1200 × 750



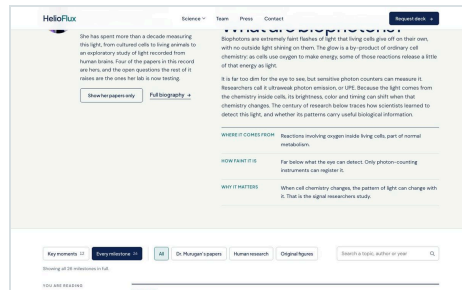
science-screen-02.jpg

1200 × 750



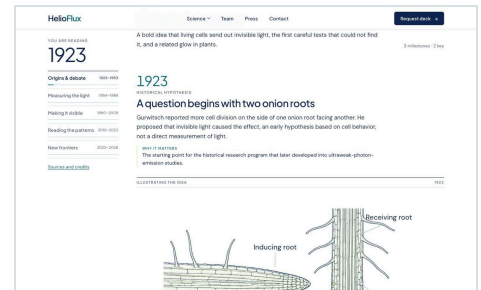
timeline-screen-00.jpg

1200 × 750



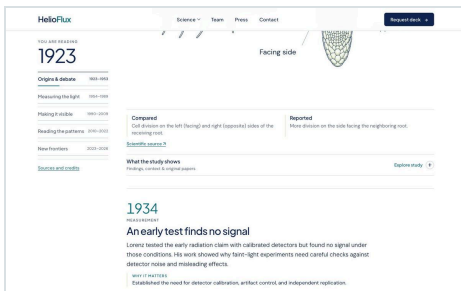
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1200 × 750



timeline-screen-02.jpg

1200 × 750



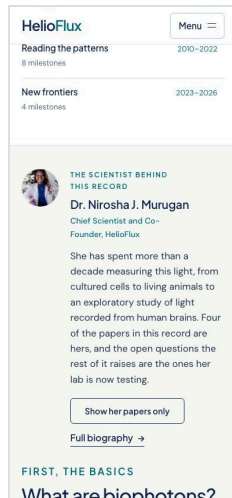
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1200 × 750



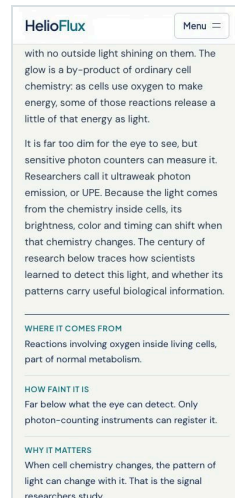
timeline-phone-00.jpg

390 × 844



timeline-phone-01.jpg

390 × 844

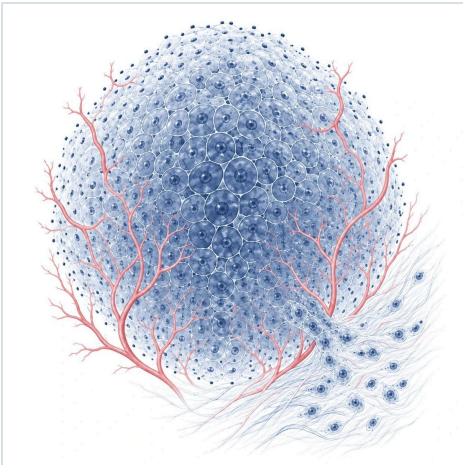


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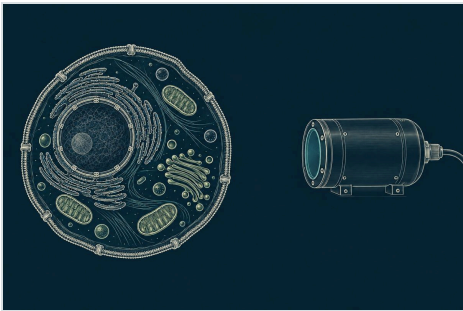
390 × 844

Plates and artwork

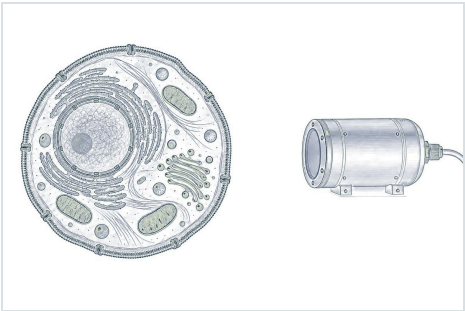
12 files: eight timeline plates, the living-light plate, the paper and night atlas, one growth frame



growth-story-frame-navy.jpg 1200 × 1200



night-cell-and-detector-hero.jpg 1200 × 800



paper-cell-and-detector.jpg 1200 × 800

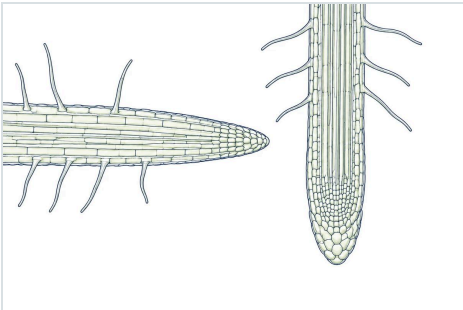


plate-1923-onion-roots.jpg 1200 × 800



plate-1934-detector-tests.jpg 1200 × 800



plate-1954-seedlings.jpg 1200 × 800

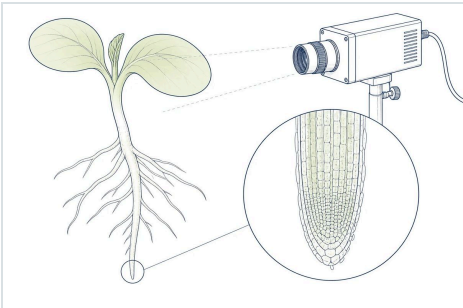


plate-1989-plant-imaging.jpg 1200 × 800

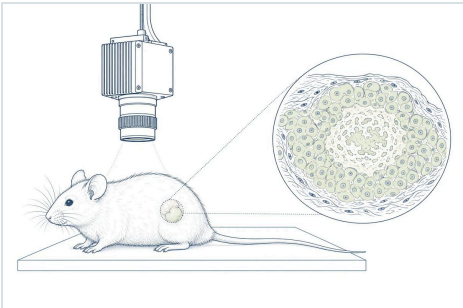


plate-2004-mouse-imaging.jpg 1200 × 800



plate-2017-oxidative-cells.jpg 1200 × 800

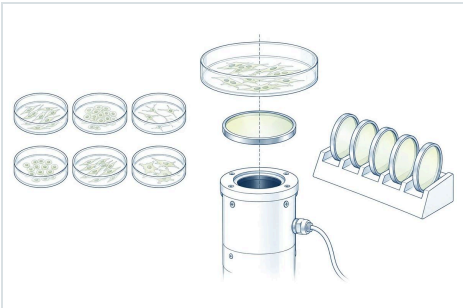


plate-2018-cell-study.jpg 1200 × 800

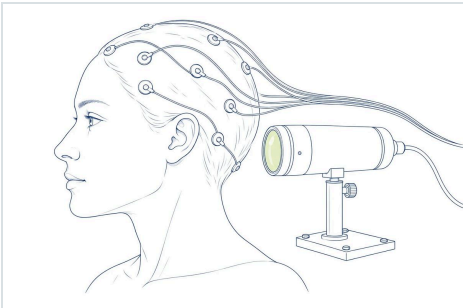


plate-2025-human-head.jpg 1200 × 800

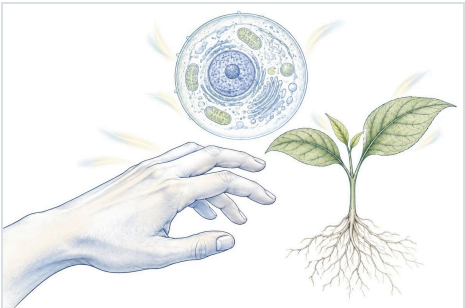


plate-living-light-hand-cell-seedling.jpg 1200 × 800